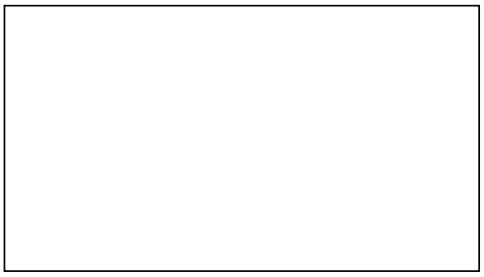
F 201: Rhino Hunt with Autopsy (15 pts + 10 extra)

What You Need for This Project

- A Windows machine with Autopsy installed



Purpose

To practice basic forensic techniques:

- Reading a scenario
- Verifying a hash value
- Extracting files from a disk image with Autopsy

Scenario

The city of New Orleans passed a law in 2004 making possession of nine or more unique rhinoceros images a serious crime. The network administrator at the University of New Orleans recently alerted police when his instance of RHINOVORE flagged illegal rhino traffic. Evidence in the case includes a computer and USB key seized from one of the University's labs. Unfortunately, the computer had no hard drive. The USB key was imaged and a copy of the dd image is the case1.zip file you've been given.

In addition to the USB key drive image, three network traces are also availableâ€”these were provided by the network administrator and involve the machine with the missing hard drive. The suspect is the primary user of this machine, who has been pursuing his Ph.D. at the University since 1972.

Downloading the Evidence File

On your Windows machine, download this file:

case1.zip (3.4 MB)

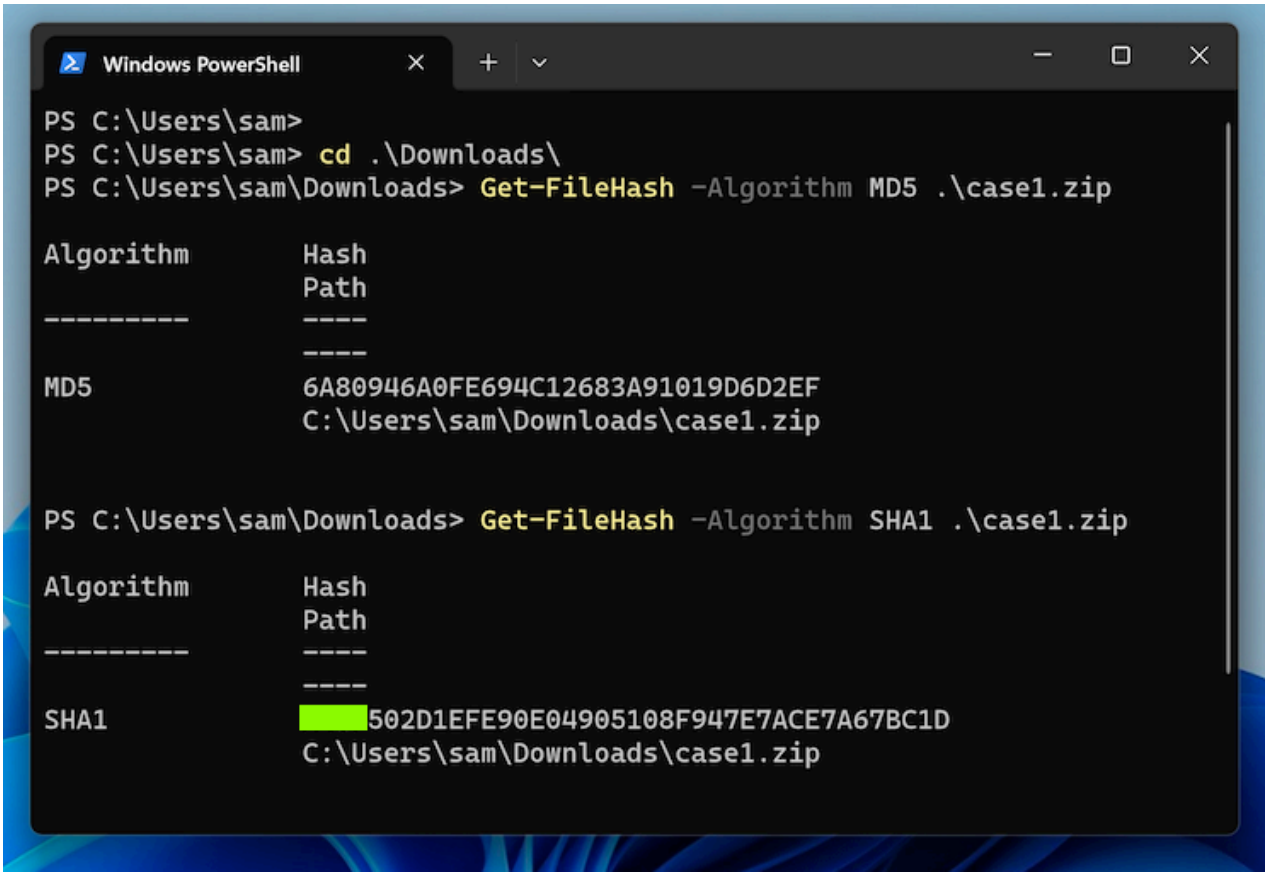
F 201.1: Verifying Hash Values (5 pts)

Open a PowerShell window and execute these commands, as shown below.

```
cd Downloads
Get-FileHash -Algorithm MD5 case1.zip
Get-FileHash -Algorithm SHA1 case1.zip
```

Verify that the MD5 value matches the value shown below. If it does not, re-download the evidence file.

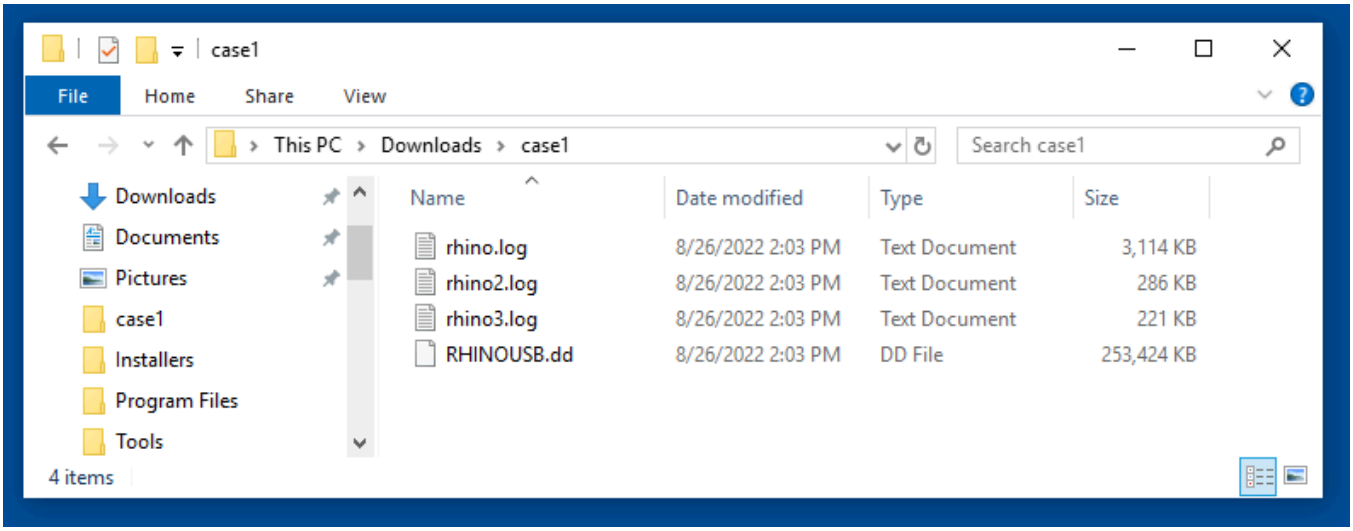
The flag is the first portion of the SHA1 address, covered by a green rectangle in the image below.



Unzipping the Evidence File

Right-click the **case1.zip** file and click "**Extract All...**". Click the **Extract** button.

You see four files, as shown below.



Creating an Autopsy Case

Launch Autopsy. In the Welcome box, click "**New Case**".

Make these selections:

- Case Name: Name your case **F201**
- Base Directory: Select your Documents folder and click **Next**.

- Assign it a case number of **F201** and click **Finish**.
- In the "1. Select Host" page, click **Next**.
- In the "2. Select Data Source Type" page, accept the default of "**Disk Image or VM File**" and click **Next**.
- In the "3. Select Data Source" page, click **Browse**, navigate to the **RHINOUSB.dd** file, and double-click it. Then click **Next**.
- In the "4. Configure Ingest" page, click the "**Select All**" button and click **Next**.
- In the "5. Add Data Source" page, Click **Finish**.

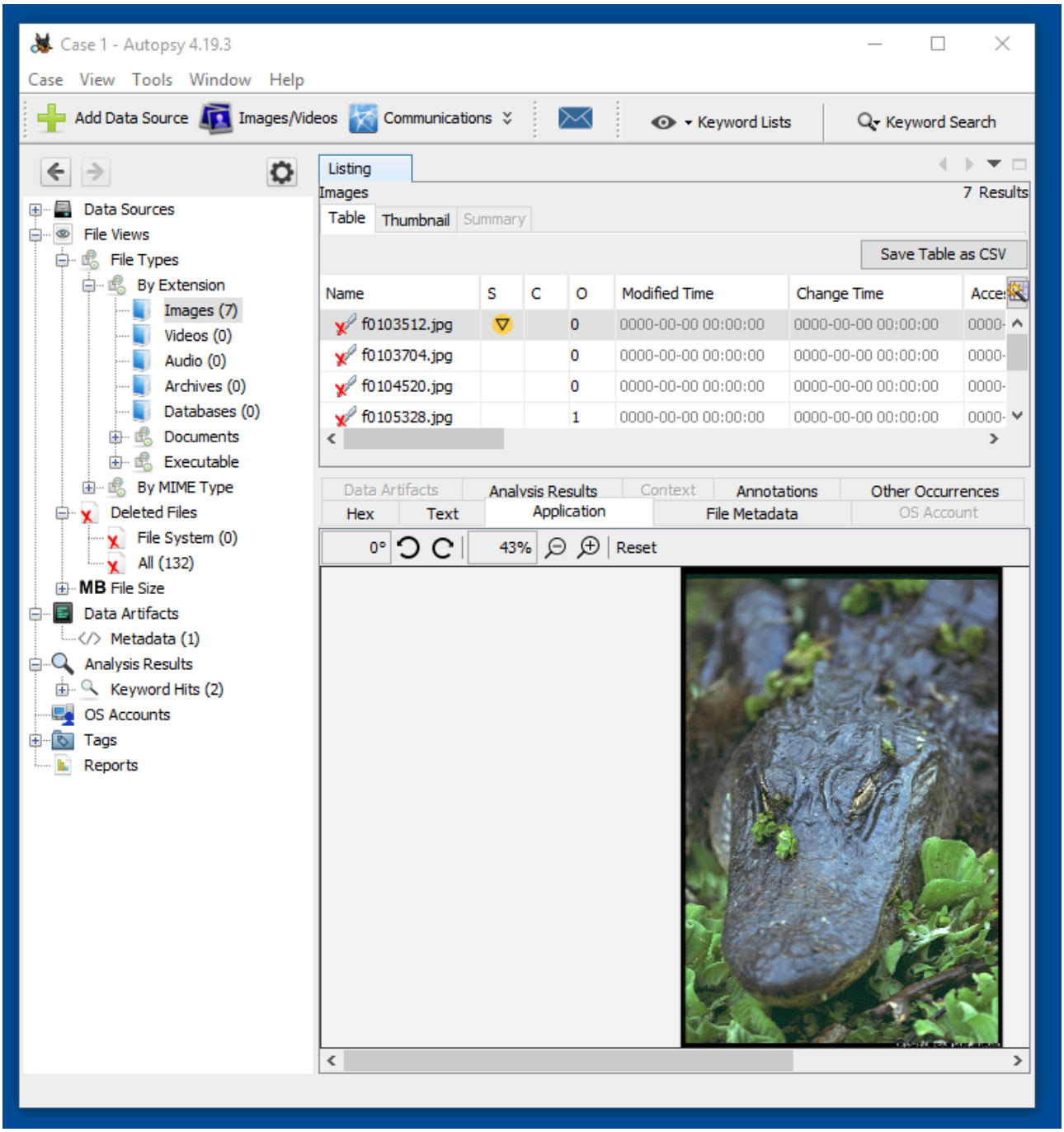
When the data file is imported and processed, in the left pane of Autopsy, expand the containers to see the **Images** and "**Deleted Files**", as shown below.

Failed Ingest Module

In Sept, 2025, the process of adding the data source failed because one of the ingest modules crashed.

If that happens, read the error message to identify the failed module.

Then repeat the process to add the data module, but uncheck the failed module in the "Configure Ingest" list.



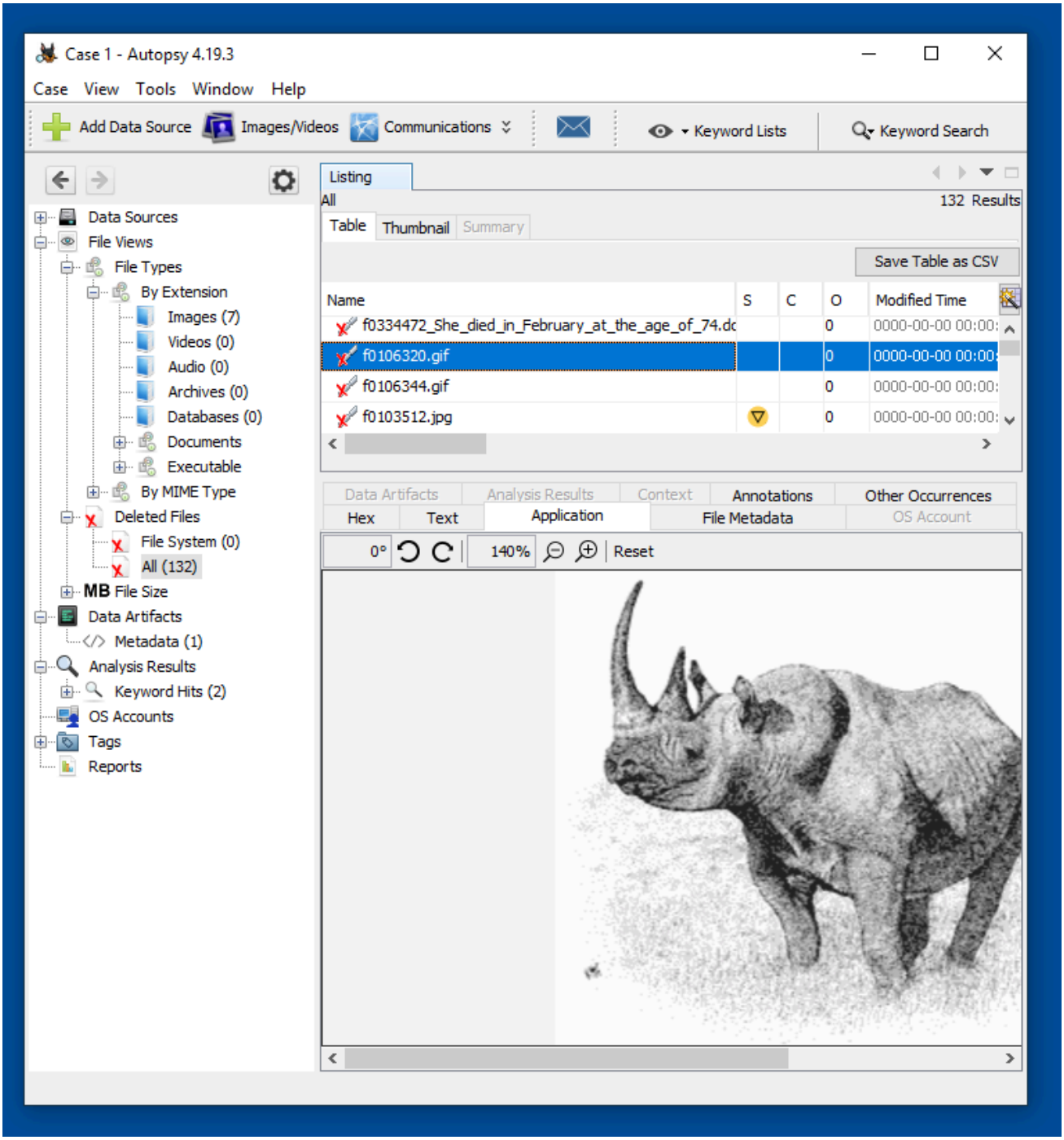
F 201.2: Mother and Child (5 pts)

Find the image of a mother rhinoceros and her child. That's the flag.

(If you are using an automated CTF scoreboard, enter the filename of the image as the flag.)

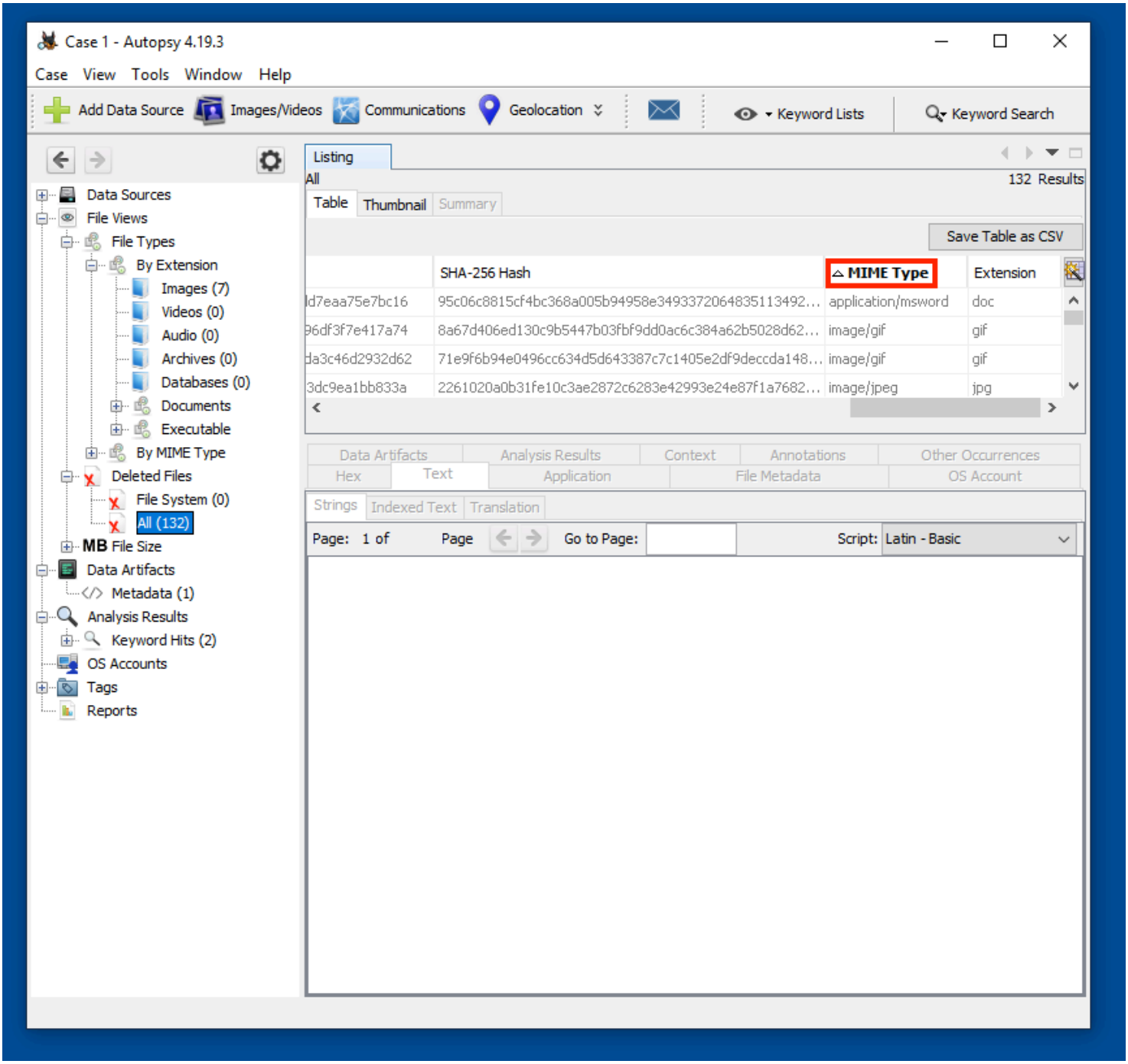
Examining Deleted Files

In the left pane, select **All**. The deleted files appear in the right pane, as shown below.



Sorting by File Type

In the top right pane, scroll to the right. Click "**MIME Type**", outlined in red in the image below, to sort the files, and put the "**application/msword**" file at the top.



Reading the Diary

The "application/msword" file is a diary. Read through it and find the flag below.

F 201.3: Hard Drive (5 pts)

Find the location of the missing hard drive. That's the flag.

F 201.4: Email Address (10 pts extra)

There are two files containing an email address at MIT. Only one of the files has a real filename. (A filename beginning with "Unalloc" is a fake filename generated by Autopsy for files recovered from

unallocated clusters.)

The flag is the real filename, which does not begin with "Unalloc".

Posted: 8-26-22
Directory selection step added 10-12-22
Video added 3-12-23
Minor correction 6-19-23
F 201.2 flag description updated 6-19-23
"Select All" added to step 4 c. 6-24-23
Flag 4 description augmented 7-15-23
Powershell Hashing info added 1-31-24
PowerShell Hashing info updated 2-6-24
Flag 4 description updated 2-13-24
Warning about failed ingest module added 9-12-25